

1. Identify rooms booked during 4-6 April:

$\text{BOOKED_ROOMS} = \pi_{\text{hotel_no}, \text{room_no}} (\sigma_{\text{bdate_from} \leq '2026-04-06' \text{ and } \text{bdate_to} \geq '2026-04-04'} (\text{BOOKING}))$

2. Get all rooms

$\text{ALL_ROOMS} = \pi_{\text{hotel_no}, \text{room_no}} (\text{ROOM})$

3. Find available

$\text{AVAIL_ROOMS} = \text{ALL_ROOMS} - \text{BOOKED_ROOMS}$

4. Join to retrieve the required attributes

$\text{R} = \pi_{\text{hotel_no}, \text{hotel_name}, \text{room_no}, \text{room_type}} (\text{AVAIL_ROOMS} \bowtie (\pi_{\text{hotel_no}, \text{room_no}, \text{room_type}} (\text{ROOM})) \bowtie (\pi_{\text{hotel_no}, \text{hotel_name}} (\text{HOTEL})))$